

ProtoDUNE HD + VD — Wire-Cell TPC Signal + Noise Simulation

one algorithm chain, two detector configurations: $\text{raw ADC}(t) = \mathcal{D}[(Q_{\text{drift}} * \text{FR} * E)(t) + n(t)]$



everything below the line is where the two configurations differ

	drift	fan-out	field response · electronics	readout window	noise spectra · ADC
PD-HD	v_d 1.565 mm/ μ s D_L 6.2 · D_T 16.3 cm ² /s	×4 APAs	dune-garfield-1d565 (APA0 own fit) single ColdElec 14 mV/fC, 2.2 μ s	6000 ticks · 500 ns	protodunehd-...-14mVfC-v1 · 14-bit
PD-VD	v_d 1.473 mm/ μ s D_L 4.0 · D_T 8.8 cm ² /s	×8 CRPs (bot 0–3 · top 4–7)	protodunevd_FR_imbalance3p_260501 bottom 7.8 mV/fC · top JSON ×1.36	6400 ticks · 500 ns	pdvd-{bottom,top}-... · 14-bit

PD-HD: [cfg/pgrapher/experiment/pdhd/sim.jsonnet](#) · see [pdhd/docs/sim-chain-diagram.md](#)

PD-VD: [cfg/pgrapher/experiment/protodunevd/sim.jsonnet](#) · see [pdvd/docs/16_pdvd-sim-chain.md](#)